

Neuron–astrocyte associative memory

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Introduction

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Lagrangians and activation functions

Associative neuron–astrocyte model

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Storing

Introduction

The paper asks whether astrocytes can contribute to associative memory.

- Classical neural network models usually focus on neurons and synapses.
- This paper adds astrocytes as an active computational component.
- The central claim is that neuron–astrocyte interactions can create higher-order memory storage.

neurons + synapses + astrocyte processes $\implies$ associative memory

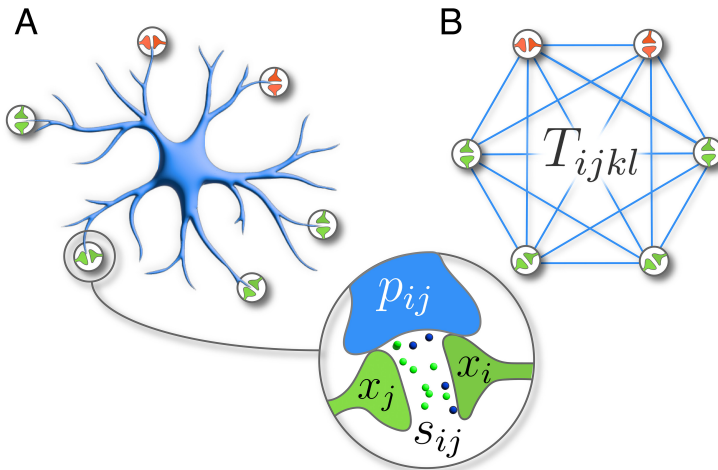
Biological motivation: tripartite synapse

- A normal synapse involves a presynaptic neuron and a postsynaptic neuron.
- Astrocyte processes can wrap around nearby synapses.
- This forms a **tripartite synapse**:

pre-neuron + post-neuron + astrocyte process.

- The astrocyte process can detect neurotransmitters, produce local Ca^{2+} signals, and release gliotransmitters.

Astrocytes can modulate synaptic transmission instead of only supporting neurons.



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synaptic activity $\rightarrow$ astrocyte $\text{Ca}^{2+} \rightarrow$ gliotransmitter release $\rightarrow$

synaptic efficacy $\rightarrow$ neural activity

- A local synapse can activate the astrocyte process surrounding it.
- Astrocyte Ca^{2+} activity can influence neurotransmitter release probability or postsynaptic response.
- This creates a feedback loop between neurons, synapses, and astrocytes.

General neuron–astrocyte dynamics

The general model contains three coupled variables:

x_i = state of neuron i ,

s_{ij} = synaptic efficacy from neuron j to neuron i ,

p_{ij} = local Ca^{2+} level in the astrocyte process around synapse $j \rightarrow i$.

The model extends a recurrent neural network by adding dynamic synapses and astrocyte calcium dynamics.

$$\tau_n \dot{x}_i = -\lambda x_i + \sum_{j=1}^N g(s_{ij}) \phi(x_j) + b_i$$

- $\dot{x}_i$ is the change rate of neuron i .
- $-\lambda x_i$ represents natural leakage or decay.
- $\phi(x_j)$ converts neuron j 's state into output activity.
- $g(s_{ij})$ is the effective synaptic weight from neuron j to neuron i .
- b_i is a baseline input.

Compared with a standard RNN, the effective weight is dynamic because it depends on s_{ij} .

$$\tau_s \dot{s}_{ij} = -\alpha s_{ij} + f(x_i, x_j, p_{ij}, s_{ij}) + c_{ij}$$

- This equation describes how the synaptic efficacy s_{ij} changes.
- It depends on the postsynaptic neuron x_i and presynaptic neuron x_j .
- It also depends on the local astrocyte Ca^{2+} level p_{ij} .
- Thus, astrocytes can indirectly regulate neuron–neuron connection strength.

Astrocytes enter the model by modulating synaptic efficacy.

$$\tau_p \dot{p}_{ij} = -\gamma p_{ij} + \sum_{k,l=1}^N T_{ijkl} \psi(p_{kl}) + \kappa(s_{ij}) + d_{ij}$$

- p_{ij} is the local Ca^{2+} level in the astrocyte process around synapse $j \rightarrow i$.
- $-\gamma p_{ij}$ represents calcium decay.
- $\sum T_{ijkl} \psi(p_{kl})$ represents influence from other astrocyte processes.
- $\kappa(s_{ij})$ represents input from the local synapse.

Astrocyte processes can connect multiple synapses through calcium signaling.

Meaning of the general model

$$x_i \leftrightarrow s_{ij} \leftrightarrow p_{ij}$$

- Neuron activity affects synaptic efficacy.
- Synaptic activity affects local astrocyte Ca^{2+} .
- Astrocyte Ca^{2+} can feed back to synaptic efficacy.
- Different astrocyte processes can also influence one another.

The model is a coupled neuron–synapse–astrocyte dynamical system.

Lagrangians and activation functions

How activation functions are defined



The paper defines activation functions from Lagrangian functions:

$$\text{activation} = \frac{\partial \mathcal{L}}{\partial z_i}.$$

Here, z can represent a neural, synaptic, or astrocytic dynamical variable.

- The Lagrangian is not the biological mechanism itself.
- It is a mathematical way to define nonlinear activation functions.
- Different Lagrangians lead to different activation functions.

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Two examples of Lagrangians

Collective activation:

$$\mathcal{L}(z) = \log \sum_{i=1}^N e^{z_i} \implies \frac{\partial \mathcal{L}}{\partial z} = \text{Softmax}(z).$$

Element-wise activation:

$$\mathcal{L}(z) = \sum_{i=1}^N Q(z_i) \implies \frac{\partial \mathcal{L}}{\partial z} = [q(z_1), \dots, q(z_N)]^T.$$

These are examples, not the only possible choices.

Associative neuron–astrocyte model

From general dynamics to associative memory

- The general model can show complicated behavior, such as oscillations or chaos.
- To study memory, the authors impose special mathematical structure.
- Under symmetry conditions, the model has a global energy function.
- The system evolves toward lower energy states.

$$\frac{dE}{dt} \leq 0$$

Stable fixed points of the energy landscape are interpreted as memories.

The paper rewrites the dynamics as a negative gradient of an energy function:

$$\tau_n \dot{x}_i = -\frac{\partial E}{\partial \phi_i} = -\lambda x_i + \sum_{j=1}^N g_{ij} \phi_j,$$

$$\tau_s \dot{s}_{ij} = -2 \frac{\partial E}{\partial g_{ij}} = -\alpha s_{ij} + \phi_i \phi_j + \psi_{ij},$$

$$\tau_p \dot{p}_{ij} = -2 \frac{\partial E}{\partial \psi_{ij}} = -\gamma p_{ij} + \sum_{k,l=1}^N T_{ijkl} \psi_{kl} + g_{ij}.$$

The negative signs mean the system moves in the direction of decreasing energy.

Why symmetry is needed

- A global energy function requires interactions to be mathematically consistent.
- This is similar to the symmetry condition in Hopfield networks:

$$g_{ij} = g_{ji}.$$

- For astrocyte processes, the corresponding condition is:

$$T_{ijkl} = T_{klij}.$$

- Intuitively, the influence from process kl to process ij must match the reverse interaction.

Symmetry prevents cyclic dynamics and allows convergence to fixed points.

The energy derivative can be written as:

$$\frac{dE}{dt} = - \left[\tau_n \sum_{i,j} \dot{x}_i \frac{\partial^2 \mathcal{L}^{[n]}}{\partial x_i \partial x_j} \dot{x}_j + \frac{\tau_s}{2} \sum_{i,j,k,l} \dot{s}_{ij} \frac{\partial^2 \mathcal{L}^{[s]}}{\partial s_{ij} \partial s_{kl}} \dot{s}_{kl} + \frac{\tau_p}{2} \sum_{i,j,k,l} \dot{p}_{ij} \frac{\partial^2 \mathcal{L}^{[p]}}{\partial p_{ij} \partial p_{kl}} \dot{p}_{kl} \right].$$

$$\frac{dE}{dt} \leq 0$$

- The Hessian terms come from differentiating the activation functions.
- If the Lagrangians are convex, these quadratic forms are nonnegative.
- The outer negative sign makes the energy non-increasing.

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The resulting effective energy is:

$$E^{\text{eff}} = \left[\sum_{i=1}^N x_i \phi_i - \mathcal{L}^{[n]} \right] - \frac{1}{4} \sum_{i,j,k,l=1}^N T_{ijkl} \phi_i \phi_j \phi_k \phi_l.$$

- The first part is the neuron-layer energy.
- The second part is a fourth-order interaction term.
- This fourth-order term is mediated by astrocyte process interactions.

The astrocyte network creates a Dense Associative Memory-like energy structure.

Storing memories

Assume there are K stored memory patterns:

$$\xi^\mu, \quad \mu = 1, \dots, K.$$

Each memory pattern is an N -dimensional vector:

$$\xi^\mu = (\xi_1^\mu, \xi_2^\mu, \dots, \xi_N^\mu).$$

- Each memory is represented as a neural activity pattern.
- The task of associative memory is to store these patterns in network weights.
- A corrupted input should converge back to the closest stored memory.

The paper stores memory patterns in the fourth-order tensor:

$$T_{ijkl} = \sum_{\mu=1}^K \xi_i^\mu \xi_j^\mu \xi_k^\mu \xi_l^\mu.$$

- Classical Hopfield networks store pairwise relations:

$$W_{ij} = \sum_{\mu} \xi_i^{\mu} \xi_j^{\mu}.$$

- This model stores fourth-order relations among neurons.
- The tensor T_{ijkl} represents higher-order interactions mediated by astrocyte processes.

Memories are stored in higher-order neuron group interactions rather than only pairwise synaptic weights.

Dense Associative Memory form

With this choice of T_{ijkl} , the effective energy becomes:

$$E^{\text{eff}} = \left[\sum_{i=1}^N x_i \phi_i - \mathcal{L}^{[n]} \right] - \sum_{\mu=1}^K F \left(\sum_{i=1}^N \xi_i^{\mu} \phi_i \right),$$

where

$$F(z) = \frac{1}{4} z^4.$$

The effective dynamics are:

$$\tau_n \dot{x}_i = -x_i + \sum_{\mu=1}^K \xi_i^{\mu} F' \left(\sum_{j=1}^N \xi_j^{\mu} \phi_j \right).$$

The network compares the current state with stored memory patterns and moves toward the most similar one.

Memory capacity

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Capacity scaling

The model contains approximately:

$$N \text{ neurons} + N^2 \text{ synapses} + N^2 \text{ astrocyte processes} \sim N^2$$

compute units.

For quartic Dense Associative Memory:

$$K^{\text{max}} \sim N^3.$$

Therefore:

$$\frac{K^{\max}}{\text{number of compute units}} \sim \frac{N^3}{N^2} \sim N.$$

The paper argues that memory capacity per compute unit grows linearly with network size.

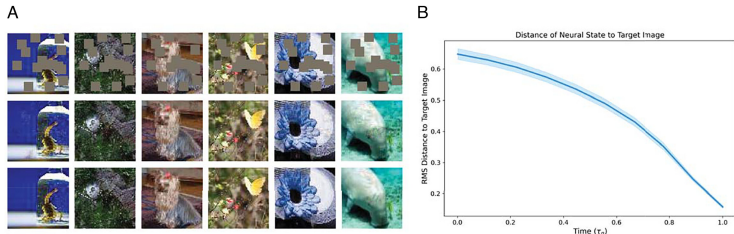


Figure: Top row is the masked input to the network, Middle row is the final state of the network, Bottom row is the ground-truth, unmasked image. Root-mean-squared distance of the state of the network to the ground-truth image, as function of time. SE was calculated across a batch of 64 images.

Interpretation of the capacity result

- Without astrocytes, memory storage mainly comes from neuron–neuron interactions.
- With astrocytes, many synapses can be connected through astrocyte processes.
- This creates more possible memory attractors in the energy landscape.
- The authors compare this favorably with other biologically plausible Dense Associative Memory implementations.

This is a theoretical capacity argument, not direct biological proof.

- The previous sections build a theoretical associative memory model.
- The simulation checks whether the model actually recovers stored memories.
- The network is given corrupted or noisy inputs.
- The dynamics should converge to the corresponding stored pattern.

corrupted input $\rightarrow$ neuron-astrocyte dynamics $\rightarrow$ retrieved memory

- The model can retrieve stored patterns from corrupted inputs.
- The energy decreases during the recovery process.
- The final fixed point corresponds to the stored memory.

The simulation supports the claim that the model can implement associative memory.

- The paper suggests that memory may not be stored only in neuron–neuron synaptic weights.
- Astrocyte processes may also contribute through calcium signaling and molecular transport.
- This provides a possible biological mechanism for higher-order memory storage.

Astrocytes may be part of the computational machinery of memory.

- The model is theoretical and idealized.
- The symmetry assumptions are mathematically useful, but real biological circuits may not satisfy them exactly.
- Future experiments could test whether disrupting astrocyte Ca^{2+} signaling impairs memory recall.
- Future models could extend from one astrocyte to larger astrocyte networks.

The paper turns astrocytes from passive support cells into active components of an associative memory model.

Astrocyte processes may link different synapses, create higher-order interactions, and increase memory storage capacity.